



PEOPLE ARE NOW EVEN DOING MACHINE LEARNING IN JAVASCRIPT

JavaScript has already emerged as the backbone of the fast-growing world of the Web. But how is open source enabling JavaScript to power advanced Web developments? **Brendan Eich**, the *creator of JavaScript*, spoke to **Jagmeet Singh** of *OSFY* to reveal the secrets of the open source framework. Eich also talked of how he enhanced the Web with **Mozilla** and the work he is doing now at **Brave Software**.

Q What was the foremost aim of developing JavaScript?

I joined Netscape to 'do Scheme' in the browser. But on my first day at work, I learned that Sun and Netscape were working on a Java integration deal. So, with Marc Andreessen directly, and Bill Joy at Sun supporting me, I came up with a plan to make a dynamic language with Java (or the C family) syntax, which people who were not professional programmers could write directly in the HTML Web page source. We wanted a scripting language to complement Java, akin to Visual Basic for Visual C++ in Microsoft's Windows platform. This would empower more people to start programming, by gluing components together with a little script in the page. The components were projected to be either built-in (the 'DOM level 0' which I implemented along with JavaScript for Netscape 2), or we hoped they could be written in Java by higher-priced programmers.

Q JavaScript is presently ubiquitous in the world of the Web. What is the reason behind its success?

There are three reasons that I see as critical and related. JavaScript itself is certainly the first and foremost reason behind its success. The second reason is its enough powerful basic features. And the third one is the ease in extending

the framework and patching (so-called 'monkey-patching' and 'object detection'). All this enabled Web developers to compensate for version differences and even extend old or incompatible browsers to resemble newer or different ones.

Q How has the open source community helped in making JavaScript the star of the Web world?

Even before Mozilla or an open source implementation of JavaScript, I used all the early adopter support techniques and energy that I had acquired over the years in software, going back to Silicon Graphics. I helped developers find workarounds and reduce test cases. I answered questions promptly. Also, I groomed helpful second and third (virtual or even real) team-mates. Netscape eventually allocated and hired more people to work on JavaScript in late 1996. Before that I had important volunteer help.

Q Do you consider JavaScript as a dominant factor in new, emerging technologies like IoT and wearables?

Thanks to Node.js and the module ecosystem it spawned, JavaScript has moved strongly into servers and IoT devices. People are now even doing machine learning in JavaScript. Early hobbyist-level work such as Johnny Five endures and grows. I expect these trends will continue over time.